

FULL NAME: _____

BLOCK: _____

Function Operations + Composition Order DOK-3 (Standalone)

Lesson 5-5 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Combine functions via $+$, $-$, $\times$, $\div$, and composition; track how domain restrictions change; recognize that composition is non-commutative and explain when the order of operations changes the outcome.
Language Objective	Justify in writing which order of a two-step discount gives a better price, using function composition.
Essential Understanding	$f \circ g \neq g \circ f$ in general — the ORDER of composition matters and often changes the answer, which matters for real-world decisions.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Add, subtract, multiply, divide, and compose functions; recognize non-commutativity of composition.
Essential Question	Why does the ORDER of a two-step discount or two-function composition change the result, and when does it not?
Materials	Student packet, pencil. DOK-3 Practice #32 (music store) is the lesson's standalone reasoning capstone — no Topic 5 assessment overlap, treated on its own merits.

[PIN] PRE-FLIGHT — STANDALONE DOK-3, NOT ASSESSED

Today's conceptual core is Practice #32 (Music Store Double Discount). Students must set up TWO different compositions: $(\$5 \text{ off}) \circ (15\% \text{ off})$ and $(15\% \text{ off}) \circ (\$5 \text{ off})$, apply to \$90, compute both, AND justify in writing which order is better for the consumer.

The justification in Part C is the DOK-3 move. Accepted answers: (a) \$72.25, (b) \$71.50, (c) 15% first is better (the \$5 saved on a smaller base is worth less than 15% off the larger base).

NOTE: Function composition is NOT on the Topic 5 assessment. Accept #32 on its own merits as a rich reasoning item.

Pace: Try-It 1 (4 min) $\rightarrow$ Try-It 4 (4 min) $\rightarrow$ #20/#21 (4 min) $\rightarrow$ #24 (3 min) $\rightarrow$ #28/#34 (4 min) $\rightarrow$ #32 (12 min). $\sim$ 35 min total Explore.

45-min Wed F variant: cut #34 and #28. Never cut #32.

Tag: lesson hook

Do Now**DOK: 1****Minutes: 5****Teacher does**

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student for profit function notation.

Students do

- Write $P(x) = R(x) - C(x)$ from the model-discuss prompt.
- Connect function subtraction to the algebraic definition.

Questions to ask

- What operation on functions gives us profit?
- Could we combine functions with other operations? (Yes — preview of today.)

Adult role

Solo teacher: circulate silently. No hints.

Do Now (5-5-savvas-model-discuss-lesson-5-5-launch)

In business, the term profit is used to describe the difference between the money the business earns (revenue) and the money the business spends (cost).

[IMAGE: Illustration of a man grooming a bulldog, with a sign reading “GROOMING \$25 per pet”.]

- A. Let x represent the number of pets groomed in a month at Grooming USA. Define a revenue function for the business.
- B. Materials and labor for each pet groomed cost \$15. The business also has fixed costs of \$1,000 each month. Define a cost function for this business.
- C. Last month, Grooming USA groomed 95 pets. Did they earn a profit? What would the profit be if the business groomed 110 pets in a month?
- D. Generalize Explain your procedure for calculating the profit for Grooming USA. Suppose you wanted to calculate the profit for several different scenarios. How could you simplify your process?

[CHECK] ANSWER KEY

Answer key (from source): A. $r(x) = 25x$. B. $c(x) = 15x + 1000$. C. No; grooming only 95 pets results in a loss of \$50 for Grooming USA. Grooming 110 pets results in a profit of \$100. D. To find the profit from 95 pets, I calculated $r(95) - c(95)$; to find the profit from 110 pets, I calculated $r(110) - c(110)$. If I needed to find the profit for several different values, it would make sense to find a function rule for profit and apply that simplified rule rather than applying two different functions and finding the difference of their values. • IMAGE PLACEHOLDER [photo]: Groomer with a dog and a sign reading “GROOMING \$25 per pet.”

Tag: teacher models Example 1 (add/subtract) + Example 4 (compose)

Launch

DOK: 2

Minutes: 12

Teacher does

- Project Example 1. Think aloud: $(f + g)(x) = f(x) + g(x)$. Domain = intersection.
- Project Example 4. Think aloud: $(f \circ g)(x) = f(g(x))$ — apply g FIRST.
- Flag the Function Operations Rules card — print it in student minds.
- 30-sec pause: “What’s the difference between $f(g(x))$ and $g(f(x))$? Check with $x = 2$.” (partner

check)

- Do NOT solve Try-Its or Practice items — release at minute 17.

Students do

- Watch Example 1 silently; copy combined-function notation.
- Watch Example 4 silently; note the inner/outer function structure.
- During pause: test $f(g(2))$ vs $g(f(2))$ with partner.
- Copy Rules card into margin.

Questions to ask

- In $(f \circ g)(x)$, which function do I apply first?
- Does $f(g(x))$ always equal $g(f(x))$? When might it?
- If $g(x) = 0$ for some x , what happens to $(f/g)(x)$ at that point?

Adult role

Project. Think aloud. Release promptly at minute 17.

[MIC] TEACHER SCRIPT

1. “Watch me add two functions. The domain of the combined function is the intersection of the two original domains.”
2. “Now composition: $(f \circ g)(x)$ means I substitute $g(x)$ ENTIRELY into f . g goes in first.”
3. “I’ll test $x = 2$: $f(g(2))$ vs $g(f(2))$. Watch what happens — the results are different.”
4. “This is the big idea today: ORDER MATTERS in composition. It’s like putting on your shoes before your socks vs. socks before shoes.”
5. “Today’s DOK-3 (#32) asks you to decide which order of a double discount is better for the consumer. You’ll set up both compositions and compare.”
6. Release to Explore.

Tag: TPS + DOK-3 driver

Explore

DOK: 2-3

Minutes: 33

Teacher does

- 5 laps across 33 min.
- Lap 1 (4 min): Try-It 1 — add/subtract. Press pairs to use function notation.
- Lap 2 (4 min): Try-It 4 — compose. Watch for inner/outer confusion.
- Lap 3 (4 min): #20/#21 — operations. Confirm domain intersections.
- Lap 4 (4 min): #24/#28/#34 — domain + evaluation. Watch for #24 domain restrictions.
- Lap 5 (12 min): [STAR] #32 Music Store. Students set up BOTH orders, compute both, and justify Part C in writing.
- Do NOT give #32 Part C answers. Pairs present argument to each other; one pair presents at Share.

Students do

- 8 items in order: Try-It 1 $\rightarrow$ Try-It 4 $\rightarrow$ #20 $\rightarrow$ #21 $\rightarrow$ #24 $\rightarrow$ #28 $\rightarrow$ #34 $\rightarrow$ #32.
- For #32: BEFORE computing, identify which function represents which discount and set up both compositions.
- Justify Part C with sentence frame.

Questions to ask

- Try-It 1: is $(f + g)(x)$ the same as $f(x + g)$? (No — watch the notation.)
- Try-It 4: which value goes in FIRST — the inner function output or x ?
- #24: what domain values are excluded, and why?
- #32: what does the \$5-off function look like? The 15%-off function?
- #32: apply each to \$90. Which gives a lower price?
- #32 Part C: WHY is one order better? (The 15% acts on a larger base amount.)

Adult role

Solo teacher: 5 laps. Press COMPOSITION ORDER justification on #32 Part C.

Try It 1 (5-5-savvas-try-it-1-lesson-5-5)

Let $f(x) = 2x^2 + 7x - 1$ and $g(x) = 3 - 2x$. Identify rules for the following functions.

a. $f + g$ b. $f - g$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $f + g = 2x^2 + 5x + 2$. b. $f - g = 2x^2 + 9x - 4$.

Try It 4 (5-5-savvas-try-it-4-lesson-5-5)

Let $f(x) = 2x - 1$ and $g(x) = 3x$. Find the value or identify the rule for the following functions.

a. $f(g(2))$ b. $f(g(x))$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $f(g(2)) = 11$. b. $f(g(x)) = 6x - 1$.

Practice #20 (5-5-savvas-q20)

Let $f(x) = (5)/(3)x^2 + 2x - (5)/(8)$ and $g(x) = 3x^2$. Identify the rules for the following functions.
SEE EXAMPLE 1

$f + g$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $f + g = (14)/(3)x^2 + 2x - (5)/(8)$.

Practice #21 (5-5-savvas-q21)

Let $f(x) = (5)/(3)x^2 + 2x - (5)/(8)$ and $g(x) = 3x^2$. Identify the rules for the following functions. SEE EXAMPLE 1

$$f - g$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $f - g = -(4)/(3)x^2 + 2x - (5)/(8)$.

Practice #24 (5-5-savvas-q24)

Let $f(x) = 4x - 5$ and $g(x) = -7x$. Evaluate each expression. SEE EXAMPLE 4

$$f(g(3))$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): -89 .

Practice #28 (5-5-savvas-q28)

Let $f(x) = x^2 + x$ and $g(x) = 9 - 2x$. Identify the rules for the following functions. SEE EXAMPLE 5

$$f \circ g$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $4x^2 - 38x + 90$.

Practice #34 (5-5-savvas-q34)

Given that $f(x) = x^2 + 8x + 3$ and $g(x) = -x - 7$, which of the following are true? Select all that apply.

- ☐ $f + g = x^2 + 7x - 4$
- ☐ $f(g(x)) = x^2 + 6x - 4$
- ☐ The domain of $(f)/(g)$ is the set of all real numbers.
- ☐ $f(x) \cdot g(x) = -x^3 - 15x^2 + 53x + 21$
- ☐ In the composition $g \circ f$, the output $f(x)$ is used as the input for g .

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): The true statements are $f + g = x^2 + 7x - 4$, $f(g(x)) = x^2 + 6x - 4$, and in the composition $g \circ f$, the output $f(x)$ is used as the input for g .

Practice #32 [STAR] DOK-3 ORDER-MATTERS — Music Store Double Discount (5-5-savvas-q32)

Use Structure A music store is running the following promotions.

[IMAGE: Promotional ad reading “HUGE SAVINGS AT O’RILEY’S MUSIC,” “\$5 off a purchase of \$20 or more,” and “Save an additional 15% off your total purchase when you open a charge account.”]

- Use composition of functions to find the sale price of a \$90 purchase when the \$5 off discount is applied prior to the 15% off discount.
- Use composition of functions to find the sale price of a \$90 purchase when the 15% off discount is applied prior to the \$5 off discount.
- In which order is the deal better for the consumer? Explain.

[STAR] DOK-3 ORDER-MATTERS — Practice #32 (standalone reasoning — NOT on Topic 5 assessment)

Students must set up TWO different compositions: $(\$5 \text{ off}) \circ (15\% \text{ off})$ and $(15\% \text{ off}) \circ (\$5 \text{ off})$, apply to \$90, compute both, AND justify in writing which order is better for the consumer.

The justification in Part C is the DOK-3 move.

Accepted answers: (a) \$72.25, (b) \$71.50, (c) 15% first is better (the \$5 saved on a smaller base is worth less than 15% off the larger base).

Anchor to Ex 4/Ex 6 (composition modeling). Sentence frame required for ELL.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. \$72.25. b. \$71.50. c. Since $\$71.50 < \72.25 , it is a better deal for the consumer if the 15% discount is applied prior to the \$5 off discount. • IMAGE PLACEHOLDER [illustration]: Promotional ad reading “HUGE SAVINGS AT O’RILEY’S MUSIC,” “\$5 off a purchase of \$20 or more,” and “Save an additional 15% off your total purchase when you open a charge account.”

Tag: academic conversation + lesson wrap

Share / Summary

DOK: 2

Minutes: 5

Teacher does

- Cold-call 1 pair to present their #32 Part C justification using sentence frame.
- Class signals thumbs. Write the composition-order definition on board: $f \circ g \neq g \circ f$.
- Lesson wrap: “Composition order matters — real decisions (discounts, unit conversions, multi-step processes) depend on it.”

Students do

- Presenting pair reads their Part C and shows both computations.
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Why does 15% off first give a better price than \$5 off first?
- Can you think of a case where the order of two functions WOULDN’T matter?

Adult role

Cold-call a pair that struggled on #32 Part C — hold them to the justification structure.

Sentence frames

Pair share (#32 Music Store): “Applying ___ first gives ___, then ___ gives ___. If I reverse the order I get ___, so $(f \circ g)(x) = ___ \neq g \circ f = ___$.”

Whole class: one pair presents their order-matters argument. Class signals thumbs up/sideways.

Tag: summary recap (not graded)

Exit Ticket

DOK: 1-2

Minutes: 3

Teacher does

- Collect at door.
- Scan stem 3 (discount order). Plan review if many students missed the direction of the inequality.

Students do

- 4 sentence stems, honest.

Questions to ask

- Did students correctly identify which order is better and give a mathematical reason?

Adult role

Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

$f \circ g$ means ___ is applied ___.

An example where $f \circ g \neq g \circ f$ is ___.

For a double discount, the better order for the consumer is ___ because ___.

One biggest thing I learned today: _____

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min. Use the extra 10 min on Explore #32 (Music Store) — the richer the written justification, the better the DOK-3 payoff.

45-min Wed F variant: cut #34 and #28 from Explore. Never cut #32.

If finished early: hand out Practice #35 (SAT/ACT $f(g(5))$) as fast-finisher.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Function Operations Rules card as a sticker/cut-out.
- For #32, provide a starter: “Let $D(x) = x - 5$ (the \$5-off function) and $P(x) = 0.85x$ (the 15%-off function). Now find $P(D(90))$ and $D(P(90))$.”
- Accept verbal Part C justification in lieu of written paragraph.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don’t skip).
- Word cards: “composition” = apply one function’s OUTPUT as the next function’s INPUT; “inner function” = applied first; “outer function” = applied second.
- “Order matters” = changing the sequence changes the answer.
- Pair ELL students with English-dominant partners for TPS.